Pulmonary Nodule Malignancy Classification and Lung Cancer Screening with Specialized 3D and 2D Deep Learning Models

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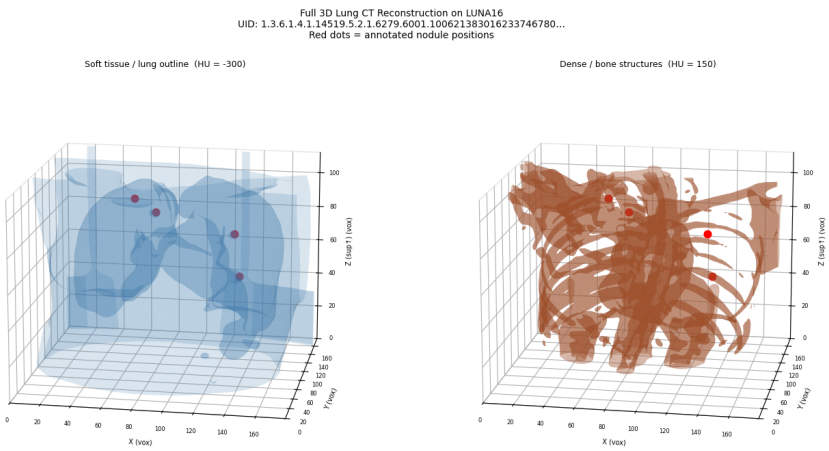


Figure 1: *Full 3D Lung CT Reconstruction on Luna16 Dataset*

ABSTRACT

Pulmonary nodule classification and lung cancer screening are related problems, but they involve different computational demands and should not automatically be treated as a single clinical task. This paper presents an iterative seven-trial study of three specialized models: *Deep3DCNN*, a 3D residual network with Squeeze-and-Excitation channel attention trained on the NoduleMNIST3D dataset for malignancy classification; *LUNA3DCNN*, a compact 3D detector trained only on the LUNA16 dataset for nodule-presence detection; and *CXRClassifier*, a fine-tuned ResNet-18 model for chest X-ray cancer screening. Each trial introduces targeted changes in architecture, training strategy, data preprocessing, and/or regularization. Across the trials, the main limitation was not model size, but interference between gradient signals from the detection and malignancy tasks when they were trained together. The final system achieves an AUROC of 0.906 on NoduleMNIST3D malignancy classification, an AUROC of 0.991 on LUNA16 nodule detection, and an AUROC 0.889 on chest X-ray cancer screening. These results show that single-task models, trained with domain-appropriate procedures, perform more reliably than shared multi-task architectures for this problem setting.

Index Terms: pulmonary nodules, lung cancer screening, computed tomography, malignancy classification, deep learning, 3D convolutional neural networks.

1. INTRODUCTION

Lung cancer is the leading cause of cancer-related mortality worldwide, accounting for approximately 1.8 million deaths each year [1]. Early detection has a major effect on survival. Five-year survival for stage I non-small cell lung cancer exceeds 80%, while stage IV survival falls below 10% [2]. For that reason, pulmonary nodule detection and malignancy classification from computed tomography, or CT, scans are very important parts of automated lung cancer screening.

These tasks operate at different visual scales. Detection, meaning whether a lesion is present, depends on larger structural features such as a dense spherical region against relatively uniform lung parenchyma [3], [4], [5]. Malignancy classification, meaning whether a confirmed nodule is benign or malignant, depends on much finer cues such as spiculation at the margins, internal density variation, lobulation, and pleural attachment [5]. When both tasks are trained together, their gradient signals do not necessarily support one another and may instead push the shared backbone in different directions [6].

This work studies three related but distinct clinical tasks, each with its own dataset and dedicated model:

1. **Deep3DCNN: Malignancy classification on NoduleMNIST3D**
   Every input is an already confirmed nodule and the model must classify it as benign or malignant from a $32 \times 32 \times 32$ CT patch.
2. **LUNA3DCNN: Nodule-Presence Detection on LUNA16**
   The model determines whether a $32 \times 32 \times 32$ patch is centered on an annotated nodule or on background lung tissue.
3. **CXRClassifier: Cancer Screening on Chest X-Ray**s
   A 2D classification model that distinguishes cancer-positive from normal radiographs in a lower-resource screening setting.

The main contributions of this work are fourfold. First, it provides a systematic seven-trial empirical study that isolates the effect of individual architectural and training changes. Second, it shows that joint multi-task training between detection and malignancy classification consistently reduces performance on both tasks because of gradient interference. Third, it introduces a two-phase transfer learning strategy, where the model is first trained on LUNA16 and then fine-tuned on NoduleMNIST3D, so that the detection task can be used for feature initialization without forcing both tasks to compete during training. Fourth, it develops a compact, task-specific 3D CNN, LUNA3DCNN, that reaches near-ceiling performance on LUNA16 by matching model complexity to task difficulty.

2. RELATED WORKS

In this section, we conduct a brief literature review on the past developments in pulmonary nodule classification, nodule detection, and chest X-ray screening, and identify the research gap addressed in this study.

2.1. 3D Convolutional Networks for Nodule Classification

The LUNA16 Grand Challenge helped establish 3D CNNs as a standard approach for pulmonary nodule detection [7]. Peters showed that classical machine learning can also perform strongly when radiologist-derived metadata are available, reporting AUROC 0.989 with a Random Forest on LIDC-IDRI clinical features [8]. This provides a strong non-deep learning baseline for nodule classification. It also highlights the effect of annotation quality, since inter-radiologist agreement on malignancy in LIDC-IDRI is only moderate, with Cohen's kappa of about 0.48 [7].

2.2. Transfer Learning and Multi-Scale Representations

Chirag compared four architectures, including a custom CNN, EfficientB0, ResNet50, and InceptionV3, on lung histopathology images and found that EfficientB0 achieved 99.01% accuracy [9]. Their results support the use of pretrained 2D backbones when the shift between source and target visual domains is not too large. The LUNA16 preprocessing notebook by Allunia describes a commonly used HU-windowing and patch extraction pipeline for 3D nodule studies and documents the Hounsfield Unit distributions that separate nodule tissue from surrounding parenchyma [10].

2.3. Chest X-ray Classification

Veyssier applied DenseNet-121 with Grad-CAM to the ChestX-ray8 dataset, which contains about 108,000 images, and reported AUROC values up to 0.90 across 14 pathologies [11]. The Grad-CAM visualizations showed that clinically meaningful anatomical regions contributed to the predictions, supporting the use of ImageNet-pretrained weights for radiograph classification. Hasan extended this comparison to Vision Transformers and found that CNN-based models with spatial attention remained

competitive on a multi-class pneumonia and COVID-19 dataset of 32,687 images [12]. Both studies also noted that differences between training and test class distributions remain a persistent issue in medical imaging, which matches the observations made in this work [11], [12].

2.4. Squeeze-and-Excitation Attention

Hu et al. introduced Squeeze-and-Excitation, or SE, networks and showed that channel-wise recalibration improved ImageNet top-1 accuracy by 0.5% to 1.0% with only a small increase in parameters [13]. SE has since been applied to 3D medical imaging, where it can be especially useful at low spatial resolutions such as $4 \times 4 \times 4$ feature maps, where spatial context is limited and more careful channel selection becomes important [14].

3. DATASETS

In this section, we describe the datasets used in this study, including their sources, class distributions, preprocessing pipelines, and roles in each task.

3.1. NoduleMNIST3D

NoduleMNIST3D is part of the MedMNIST v2 benchmark and is derived from the LUNA16 challenge dataset [15]. Each sample is a $28 \times 28 \times 28$ voxel patch centred on a radiologist-annotated pulmonary nodule and preprocessed to uint8 using the MedMNIST normalization pipeline. Labels are binary, with benign assigned 0 and malignant assigned 1. The dataset contains 1,158 training samples, 165 validation samples, and 310 test samples. The class distribution is roughly 3 to 1 benign to malignant, which creates a meaningful imbalance with a natural positive rate of about 25%. Because every sample is already a confirmed nodule, the model is not learning to detect nodules. It must instead distinguish finer differences in texture and shape. This dataset is the main benchmark used to evaluate Deep3DCNN across all trials.

From Trial 6 onward, all experiments upsample the patches to $32 \times 32 \times 32$ using trilinear interpolation in the data pipeline. This gives the model one extra MaxPool stage and yields $4 \times 4 \times 4$ feature maps before global average pooling, rather than $3 \times 3 \times 3$ at the original 28 cubed resolution.

3.2. LUNA16

The Lung Nodule Analysis 2016, or LUNA16, challenge provides 888 CT scans from the LIDC-IDRI collection with precise 3D centroid annotations from multiple radiologists [7]. Positive patches are extracted at annotated nodule centroids, while negative patches are sampled from background lung regions with no nodule within 6 mm. The task is binary detection, meaning determining if a nodule is present or it is background tissue. Unlike NoduleMNIST3D, the positive and negative classes are visually distinct. A dense spherical structure can usually be separated from uniform lung tissue even at $32 \times 32 \times 32$ resolution.

Patches are extracted using HU windowing from -1000 to 400, followed by percentile-based normalization from the 1st and 99th percentiles to the range 0 to 255. This was chosen to better match MedMNIST-style normalization and reduce the preprocessing mismatch that emerged as an important source of domain shift in Trial 4. The dataset contains approximately 615 positive and 1,335 negative training samples, split 80, 10, 10 in a stratified way by scan. In the final system, LUNA16 is used only for LUNA3DCNN training and evaluation.

3.3. Lung Cancer X-Ray

The chest X-ray screening dataset used in this work was obtained from the Kaggle dataset Lung cancer x-ray [16]. It is a binary dataset with two classes, Cancer and NORMAL. The training folder contains 5,216 images. A stratified 15% internal validation split, corresponding to 783 images, was taken from the training folder. The held-out test set contains 624 images. Images were resized to 224×224 and normalized with mean 0.5 and standard deviation 0.5. Training augmentation included random horizontal flips, rotations of up to 10°, affine shear, and colour jitter. The difference between the training and test class distributions motivated the use of temperature scaling and threshold calibration.

3.4. Supplementary Datasets Used in Trials 3 and 4

Two additional 2D datasets were used in Trials 3 and 4: IQ-OTH:NCCD, with 877 training

JPEG slices, and LungcancerDataSet, with 1,460 training JPG or PNG slices [17], [16]. Both were converted into pseudo-3D volumes by repeating each 2D slice 28 times along the depth axis. This increased the amount of training data, but it did not add real depth information. As a result, the 3D convolutional kernels received no useful signal along the stacked axis. Both datasets were effectively saturated, with AUROC values of at least 0.997 from Trial 3 onward, and they were removed after Trial 4.

4. METHODS

In this section, we present the model architectures, training strategies, preprocessing steps, and evaluation procedures used throughout the study.

4.1. Architecture Overview

4.1.1. Small3DCNN

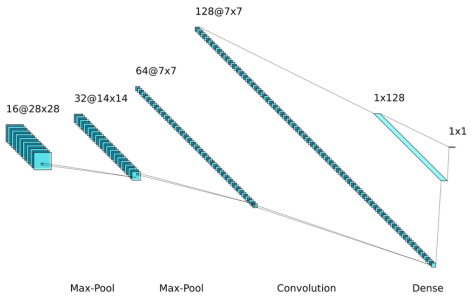


Figure 2: *Small3DCNN Model Architecture*

Used in Trials 0 to 3, Small3DCNN serves as the baseline model. The motivation for this architecture was to provide a compact and interpretable starting point for the study, allowing later trials to measure the effect of added capacity, residual connections, and training changes against a fixed reference model. Its relatively small size also made it appropriate for establishing whether the NoduleMNIST3D patches contained enough discriminative 3D information for malignancy classification before moving to larger models. It contains two residual blocks introduced in Trial 2, channel widths of 16 to 32 to 64 to 128, three convolutional stages, and a Global Average Pooling to Dropout of 0.3 to FC 128 to 1 head. The total parameter count is approximately 884,000.

4.1.2. Deep3DCNN

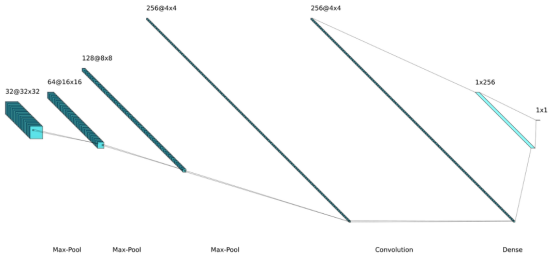


Figure 3: *Deep3DCNN Model Architecture*

Used in Trials 4 to 7, Deep3DCNN is a wider and deeper architecture with four residual stages at 32, 64, 128, and 256 channels, and three MaxPool layers that reduce spatial resolution from 32 cubed to 4 cubed. The motivation for this model was that NoduleMNIST3D is the primary malignancy-classification benchmark in this study, and this task depends on subtle morphological cues such as spiculation, lobulation, and internal density variation. A higher-capacity 3D network was therefore needed to capture richer volumetric structure than the smaller baseline could represent. From Trial 7 onward, each ResBlock3D is replaced with SEResBlock3D, which adds Squeeze-and-Excitation channel attention. This was motivated by the fact that the most informative signal may vary from case to case, with some nodules being distinguished more by boundary texture, others by shape, and others by internal density patterns. The classifier head is Global Average Pooling to Dropout 0.5 to FC 256 to 1. The total parameter count is about 3.7 million.

The SE module performs channel-wise recalibration. The spatial dimensions of each feature map are globally average pooled to produce a batch by channel descriptor. This passes through two linear layers, FC from C to C divided by 8, followed by ReLU, then FC from C divided by 8 back to C, followed by Sigmoid. The result is a set of per-channel weights between 0 and 1 that rescale the original feature map. At the $4 \times 4 \times 4$ stage, where spatial context is limited, this helps the network emphasize channels that capture malignancy-relevant patterns such as spiculation and density gradients, while reducing the effect of noisier channels.

4.1.3. LUNA3DCNN

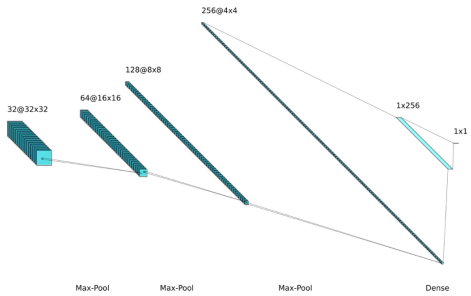


Figure 4: *Luna3DCNN Model Architecture*

Introduced in Trial 7, LUNA3DCNN is a compact 3D CNN built specifically for LUNA16 nodule detection. The motivation for this model was that LUNA16 and NoduleMNIST3D, although both CT-based, represent different tasks. LUNA16 is a nodule-presence detection problem with a cleaner decision boundary, while NoduleMNIST3D is a malignancy-classification problem requiring finer discrimination. A separate model was therefore used to avoid the domain and calibration conflict seen when both tasks were trained together. Because LUNA16 detection depends more on larger structural differences than on subtle channel-level cues, a simpler dedicated architecture was more appropriate than reusing the full Deep3DCNN with SE attention. Its architecture is Stem 32, then ResBlock3D from 32 to 64 with pooling, ResBlock3D from 64 to 128 with pooling, ResBlock3D from 128 to 256 with pooling, followed by Global Average Pooling, Dropout 0.3, and FC 256 to 1. The total parameter count is about 2.0 million. SE attention is deliberately excluded because the detection task is visually simpler and depends more on large-scale structural differences than on subtle channel-level distinctions. The lower dropout reflects the simpler decision boundary.

4.1.4. CXRClassifier

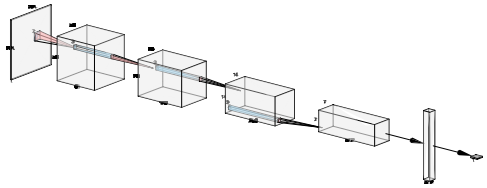


Figure 5: *CXRClassifer Model Architecture*

Used in Trials 5 to 7, CXRClassifier is a pretrained ResNet-18 backbone adapted for grayscale input by averaging the original RGB first-convolution weights across channels. The motivation for this model was that the chest X-ray task is fundamentally two-dimensional and should not be forced into a 3D CNN framework. Unlike CT-based nodule classification, chest radiograph screening depends on 2D texture and shape patterns, making a pretrained 2D backbone a better match for the data. This model also fills a complementary role in the overall screening pipeline, since chest X-rays are more widely available than CT scans and can serve as an earlier, lower-cost screening stage before CT-based follow-up. Its head is Global Average Pooling to Dropout 0.5 to FC 512 to 1. The total parameter count is about 11.2 million. Differential learning rates are used, with 1e-4 for the backbone and 5e-4 for the classification head. After training, probability outputs are calibrated using post-hoc temperature scaling.

4.2. Training Strategies

4.2.1. Loss Functions

All models use binary cross-entropy with label smoothing of epsilon 0.05. This changes hard targets of 0 and 1 into $\frac{\epsilon}{2}$ and $1 - \frac{\epsilon}{2}$ respectively, which helps prevent overconfident logits and improves calibration under noisy medical labels. A positive class weight equal to the negative to positive class ratio is also used to increase gradient signal for the minority class.

4.2.2. Class Imbalance Handling

NoduleMNIST3D has an approximate 3 to 1 benign to malignant ratio. From Trial 7 onward, a WeightedRandomSampler oversamples malignant cases to a target of 40% per batch so that the minority class contributes more consistently during training. This is used together with pos_weight rather than replacing it.

4.2.3. Two-Phase Training

In Trial 7, Deep3DCNN was trained in two phases. Phase 1 trains the model on LUNA16 for 20 epochs using a learning rate of 3e-4 and CosineAnnealingLR. The goal is to initialize the backbone with general 3D volumetric features such as HU boundary detectors, spherical density patterns, and parenchyma texture filters. Phase 2 fine-tunes the network on NoduleMNIST3D for up to 60 epochs

using a learning rate of 1e-4, a 5-epoch linear warmup, cosine decay to 1e-6, and patience of 10 on validation AUROC. Gradient clipping with max norm 1.0 is applied in both phases.

4.2.4. 3D Augmentation

All 3D models use a custom collate function for augmentation. This includes random flips along each axis with 50% probability, random 90-degree rotations in any axis pair with 50% probability, mild zoom-crop with probability 30% and scale from 1.0 to 1.15, multiplicative intensity jitter with probability 40% and factor from 0.85 to 1.15, additive Gaussian noise with probability 40% and sigma 0.05, and random 3D cutout with probability 20%, where an $8 \times 8 \times 8$ cube is zeroed out.

4.2.5. CXRClassifier Temperature Scaling

After CXRClassifier training, the temperature scalar T is optimized on validation logits to minimize binary cross-entropy negative log-likelihood, where the calibrated probability is defined as:

$$\sigma\left(\frac{logit}{T}\right)$$

This represents the sigmoid of logit divided by T. A value of T greater than 1 softens an overconfident model, while a value below 1 sharpens an underconfident one. All threshold sweeps and test evaluations use calibrated logits.

4.2.6. Threshold Tuning

The classification threshold is tuned by sweeping values from 0.05 to 0.95 on the validation set and selecting the value that maximizes F1 score. Each model is tuned on its own validation set so that threshold selection stays tied to the domain on which the model is being used.

5. RESULTS

In this section, we report the experimental results across all trials and compare model performance on the malignancy classification, nodule detection, and chest X-ray screening tasks. Table 1 summarizes Deep3DCNN performance on the fixed NoduleMNIST3D test set across all trials, which serves as the main cross-trial benchmark as the test set remains unchanged throughout the study.

5.1. Trial by Trial Analysis

5.1.1. Trials 0 to 2: Architecture Foundations

Trials 0 to 2 established a strong baseline for the NoduleMNIST3D task. Trial 0 began with the Small3DCNN model and achieved an AUROC 0.849, already above the logistic regression baseline of 0.823.

Trial 1 improved this to 0.871 after adding learning-rate scheduling and threshold tuning, which showed that part of the early performance gain came from better optimization and a better-calibrated operating point.

Trial 2 introduced residual connections through ResBlock3D and increased AUROC further to 0.894, indicating that the deeper residual formulation helped the network preserve and refine useful 3D features.

5.1.2. Trial 3: Best AUROC Through Expanded Data

Trial 3 produced the highest NoduleMNIST3D AUROC in the study at 0.922. This trial added the pseudo-3D supplementary datasets

Table 1: *Iterative Trial Summary on the NoduleMNIST3D Benchmark*

Trial	Key Change	AUROC	F1	Recall	Precision	Thr
0	Baseline Small3DCNN, fixed thr	0.849	0.564	0.516	0.623	0.50
1	LR scheduling & threshold tuning	0.871	0.598	0.594	0.603	0.65
2	ResBlock3D & val loss tracking	0.894	0.627	0.734	0.547	0.50
3	Expanded data & augmentation	**0.922**	0.678	0.641	**0.719**	0.60
4	Deep3DCNN & LUNA16 (joint)	0.830	0.584	0.625	0.548	0.85
5	Dual-model & LUNA16 alignment	0.847	0.592	**0.828**	0.461	0.55
6	NoduleMNIST3D-only & 32^3 patches	0.915	0.671	0.766	0.598	0.45
7	SE attention & two-phase training	0.906	**0.700**	0.766	0.645	0.85

together with stronger augmentation, and the result was a clear improvement in score separation. Precision rose to 0.719 and the tuned threshold moved to 0.60, suggesting that the model became more selective on benign cases while still maintaining competitive recall. At this stage, the added data appeared beneficial on the main benchmark

5.1.3. Trial 4: Joint Multi-Domain Training Hurt the Main Benchmark

Trial 4 replaced the smaller network with Deep3DCNN and introduced joint training with LUNA16. Although this increased model capacity and added genuine 3D CT data, NoduleMNIST3D AUROC fell sharply to 0.830. This was the largest drop across the full sequence of trials. The result suggests that the shared training objective pulled the backbone toward features that were useful for nodule detection but less effective for benign-versus-malignant discrimination. In other words, the model became better at recognizing the presence of a nodule, but worse at learning the finer cues required for malignancy classification.

5.1.4. Trial 5: Dual-Model Setup and LUNA16 Preprocessing Alignment

Trial 5 partially recovered NoduleMNIST3D AUROC to 0.847 after removing the pseudo-3D datasets and aligning the LUNA16 preprocessing more closely with the MedMNIST pipeline. As shown in Fig. 6, both training and validation loss dropped quickly during the early epochs and then stabilized, with the best checkpoint selected at epoch 14. The gap between the two curves remained modest, so severe overfitting was

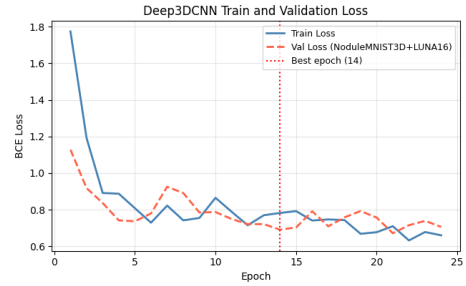


Figure 6: *Deep3DCNN Training and Validation Loss*

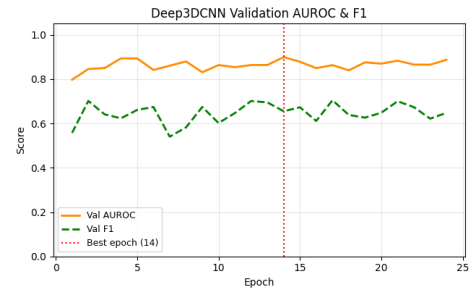


Figure 7: *Deep3DCNN Train and Validation Loss*

not the main issue. Fig. 7 shows that validation AUROC remained stable in the high 0.80s, while validation F1 fluctuated more noticeably across epochs.

This suggests that the model learned a reasonably stable shared representation, but the classification boundary remained sensitive to threshold choice and class balance. Trial 5, therefore, improved optimization and preprocessing consistency, but it did not resolve the underlying conflict between the LUNA16 detection task and the NoduleMNIST3D malignancy task.

Final Results Summary for All Models

Run: Trial07_2026-04-14_18.18.57 | Gold cells = best value per metric

Model	Dataset	AUROC	F1	Recall	Precision	Specificity	Accuracy	Thr
Deep3DCNN (SE Attention)	NoduleMNIST3D	0.906	0.700	0.766	0.645	0.890	0.865	0.85
LUNA3DCNN	LUNA16	**0.991**	**0.934**	0.934	**0.934**	**0.970**	**0.959**	0.55
CXRClassifier (T* = 0.503)	Chest X-ray	0.889	0.883	**0.997**	0.792	0.564	0.835	0.25
LogReg (baseline)	NoduleMNIST3D	0.823	0.538	0.547	0.530	0.874	0.806	--

Deep3DCNN (NoduleMNIST3D) | LUNA3DCNN (LUNA16) | CXRClassifier (Chest X-ray) | LogReg baseline | Best value per metric

Figure 8: *Final Results Summary for All Models*

5.1.5. Trial 6: NoduleMNIST3D Specialization

In Trial 6, we removed LUNA16 from Deep3DCNN training and returned the model to a single-task setting on NoduleMNIST3D, while also using $32 \times 32 \times 32$ input patches. This raised AUROC to 0.915, which was a major recovery from Trial 5 and the clearest evidence that joint training had been hurting the main benchmark. The result supports the view that the earlier performance drop was caused primarily by task interference rather than by insufficient model capacity. The higher recall of 0.766 also showed that the model became better at recovering malignant nodules once the training objective was narrowed again.

5.1.6. Trial 7: Three Dedicated Specialists

Trial 7 finalized the system as three dedicated specialists: Deep3DCNN for NoduleMNIST3D malignancy classification, LUNA3DCNN for LUNA16 nodule detection, and CXRClassifier for chest X-ray screening. This design allows each model to be tuned on its own domain and with its own decision threshold, rather than forcing a shared backbone or a shared operating point across tasks with different score distributions and different error trade-offs.

5.2. Final System Results

5.2.1. Deep3DCNN on NoduleMNIST3D Malignancy Classification

The final Deep3DCNN was tuned on the NoduleMNIST3D validation set with a threshold of 0.85. As shown in Fig. 9, F1 increases steadily as the threshold rises and reaches its best region in the high-threshold range, while recall decreases and specificity increases. This indicates that the final model performs best when the decision boundary is shifted upward rather than left near the midpoint. The corresponding ROC curve in Fig. 10 reports validation AUROC of 0.899 and places the chosen operating point at a low false-positive rate with a moderate true-positive rate.

On the test set, the tuned-threshold confusion matrix in Fig. 19 contains 219 true negatives, 27 false positives, 15 false negatives, and 49 true positives. These values correspond to F1 0.700, recall 0.766, precision 0.645, specificity 0.890, and accuracy 0.865. In practice, this means the final malignancy model operates conservatively as it gives up some sensitivity relative to a lower threshold, but it keeps false positives under better control and produces a more balanced trade-off between recall and specificity.

5.2.2. Luna3DCNN on Luna16 nodule detection

LUNA3DCNN showed the cleanest training behaviour of the three models. In Fig. 11, both training and validation loss decrease steadily across training, with the best checkpoint selected at epoch 46. Fig. 12 shows that validation AUROC rises quickly during the early epochs and then stabilizes near 0.98 to 0.99, which suggests that the model continued to benefit from longer training without becoming unstable. The threshold analysis in Fig. 13 shows that performance remains strong across a fairly wide threshold range, with the selected threshold of 0.55 sitting near a balanced operating point. The ROC curve in Fig. 14 reports validation AUROC 0.988.

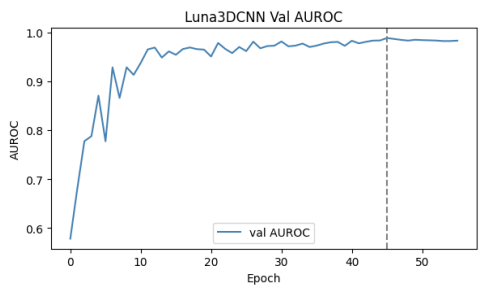


Figure 9: *Deep3DCNN Threshold Analysis*

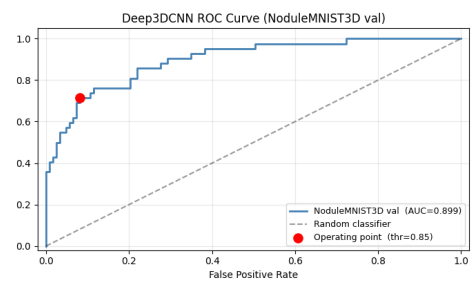


Figure 10: *Deep3DCNN ROC Curve*

On the test set, the confusion matrix in Fig. 19 contains 130 true negatives, 4 false positives, 4 false negatives, and 57 true positives. These values give F1 0.934, recall 0.934, precision 0.934, specificity 0.970, and accuracy 0.959. The matched precision and recall are especially useful here, because

they show that the selected threshold did not heavily favour one error type over the other. Overall, the results indicate that nodule detection on LUNA16 is a much cleaner classification problem than malignancy grading, and that a dedicated compact 3D model is sufficient to solve it well.

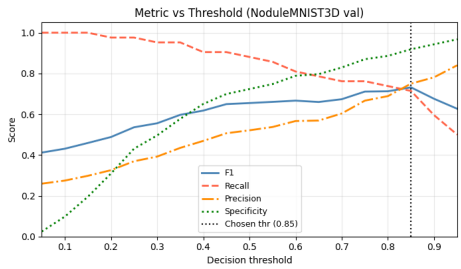


Figure 11: *Luna3DCNN Train and Validation Loss*

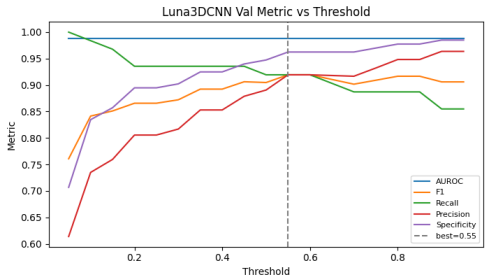


Figure 12: *Luna3DCNN Validation AUROC*

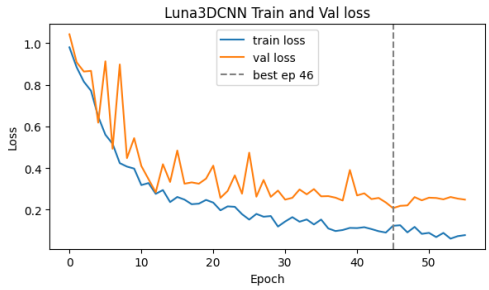


Figure 13: *Luna3DCNN Threshold Analysis*

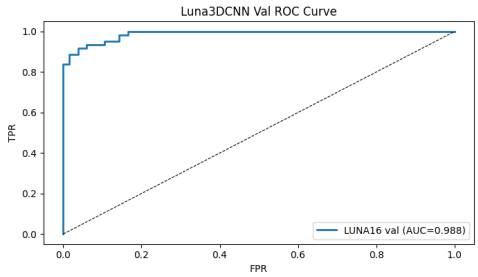


Figure 14: *Luna3DCNN ROC Curve*

5.2.3. CXRClassifier on Chest X-Ray Cancer Screening

CXRClassifier also converged quickly. In Fig. 15, training and validation loss remain close throughout the run, with the best checkpoint selected at epoch 21. Fig. 16 shows that validation AUROC and F1 stay near the ceiling for most of the training, which indicates that the model learned a strong class boundary early and then mainly refined it. After temperature scaling, the threshold analysis in Fig. 17 shows that the chosen threshold of 0.25 lies in a region where recall remains almost maximal across a wide range of thresholds, while specificity improves only modestly as the threshold increases. The calibrated ROC curve in Fig. 18 reports validation AUROC 0.999.

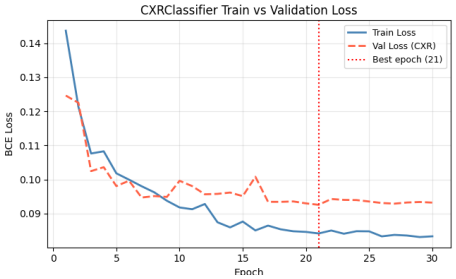


Figure 15: *CXRClassifier Train and Validation Loss*

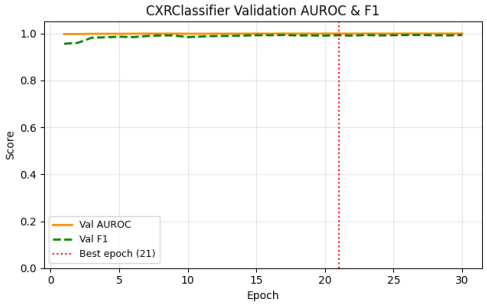
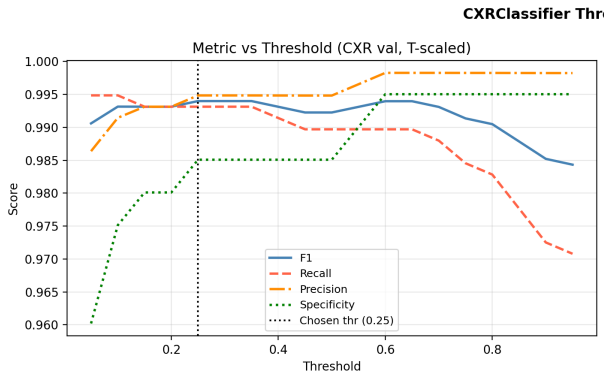


Figure 16: *CXRClassifier Validation AUROC and F1*

On the test set, the confusion matrix in Fig. 19 contains 132 true negatives, 102 false positives, 1 false negative, and 389 true positives. These values correspond to F1 0.883, recall 0.997, precision 0.792, specificity 0.564, and accuracy 0.835. The model therefore behaves like a screening tool as it misses almost no cancer cases, but it does so by accepting a substantial number of false positives. For an initial screening stage, this is a reasonable trade off, since flagged cases can be referred for follow-up imaging rather than dismissed.



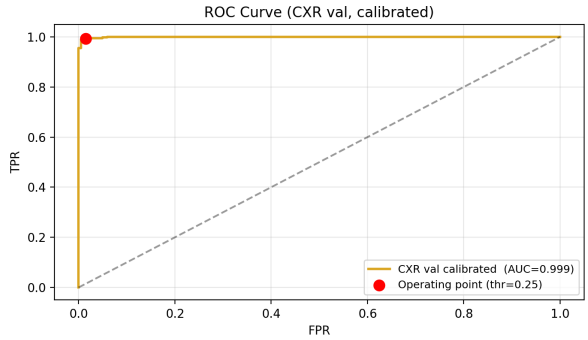


Figure 17 (Left): *CXRClassifier Threshold Analysis*

Figure 18 (Right): *CXRClassifier ROC Curve*

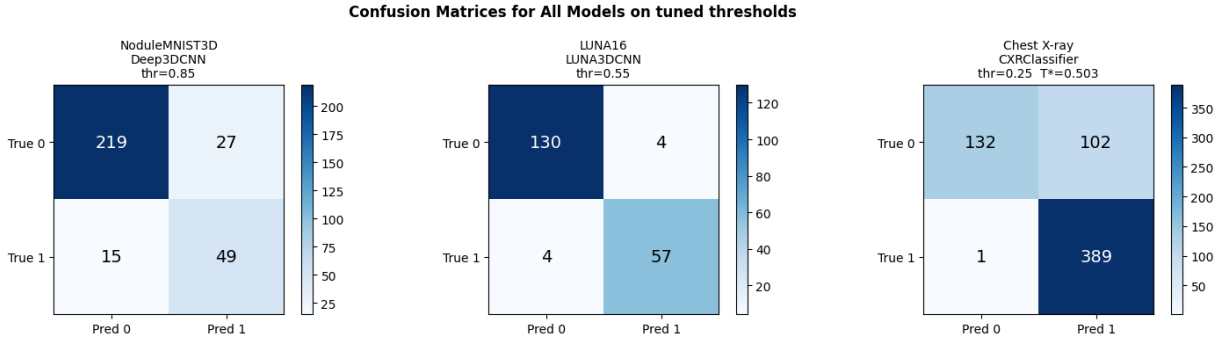


Figure 19: *Confusion Matrices for Models: Deep3DCNN, Luna3DCNN, & CXRClassifier*

5.3. Visualization

Fig. 1 shown on page 1, illustrates a full 3D lung CT reconstruction from LUNA16 with annotated nodule locations overlaid. This figure is useful because it places the detection problem in its anatomical context. The nodules are embedded within the larger lung volume, and the reconstruction shows that the detection task involves locating compact structures within surrounding lung tissue rather than deciding whether an already-centered patch is malignant. This helps explain why LUNA16 detection is an easier and more spatially distinct task than NoduleMNIST3D malignancy classification.

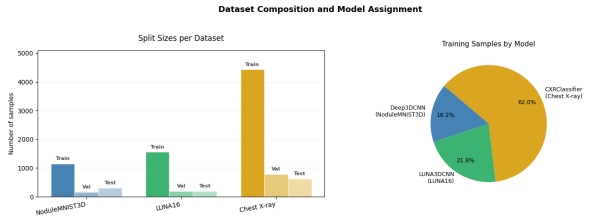


Figure 20: *Dataset Composition and Model Assignment*

Beyond the numerical results, the visualizations provide useful context for the data distributions, the task differences, and the internal behaviour of the models. Fig. 20 summarizes dataset composition and model assignment in the final system. It shows that the three datasets were not used interchangeably, but were instead mapped to separate specialist models according to task and modality. NoduleMNIST3D was used for malignancy classification with Deep3DCNN, LUNA16 was used for nodule detection with LUNA3DCNN, and the chest X-ray dataset was used for radiograph screening with CXRClassifier. This figure helps explain the final system design, since the best results were obtained only after separating the tasks rather than forcing them into a shared training pipeline.

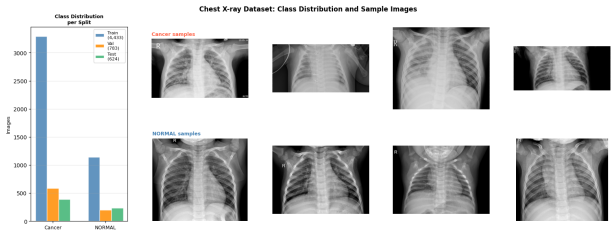


Figure 21: *Chest X-Ray Class Distribution and Sample Images*

Fig. 21 shows the class distribution and representative chest X-ray images. The class counts make the train, validation, and test composition immediately visible, while the example radiographs illustrate the visual variability within both the Cancer and NORMAL classes. This is useful context for interpreting the screening results, particularly because the chest X-ray classifier was optimized for very high recall and therefore operated in a regime where false positives were acceptable. The sample images also highlight that this task differs substantially from the CT-based nodule tasks in both modality and visual scale.

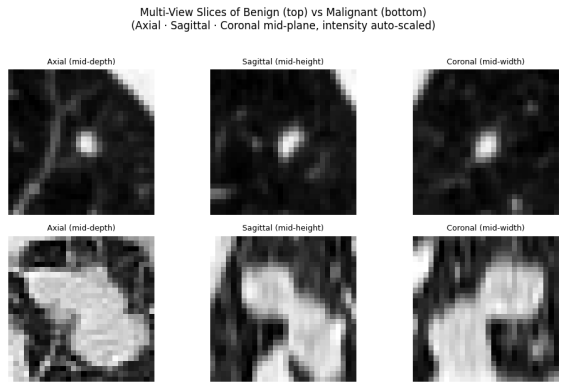


Figure 22: *Multi-View Slices of Benign and Malignant Nodules*

For the malignancy-classification task, Fig. 22 compares axial, sagittal, and coronal mid-plane slices for benign and malignant nodules. Viewing the nodules across all three anatomical planes makes the difference between detection and malignancy grading more concrete. The benign example appears relatively compact and localized, whereas the malignant example shows a larger and less regular structure across views. These slice comparisons help illustrate why malignancy classification is more difficult than simple nodule detection, since the model must respond to subtle differences in internal structure and boundary appearance rather than only to the presence of a dense object [18].

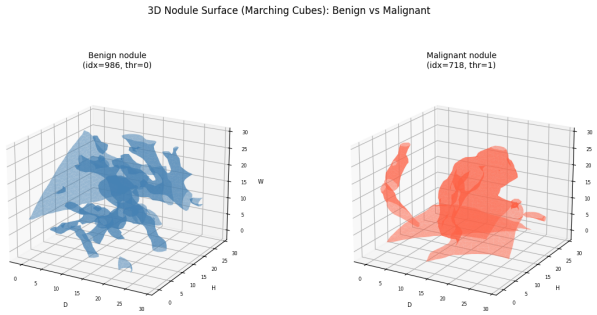


Figure 23: *Benign vs Malignant 3D Nodule Surface*

Fig. 23 shows 3D surface reconstructions of a benign and a malignant nodule using marching cubes after smoothing and thresholding. These surfaces provide a qualitative view of morphological differences between the two examples. Under the same reconstruction settings, the malignant example appears larger and more consolidated, while the benign example appears smaller and more fragmented. Although these reconstructions are only illustrative and should not be treated as direct clinical measurements, they help reinforce the main challenge of the malignancy task: the distinction depends on fine structural differences in shape and internal organization, not simply on whether a nodule is present.

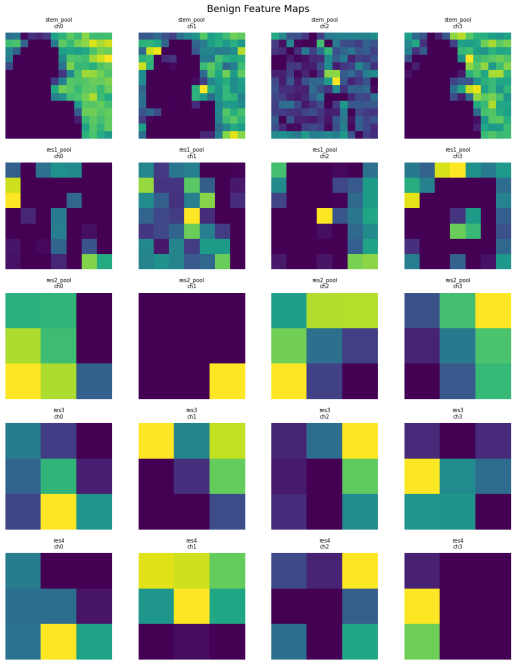


Figure 24: *Intermediate Feature Maps for a Benign Nodule from Deep3DCNN*

Finally, Fig. 24 presents intermediate feature maps for a benign nodule from Deep3DCNN. The early stages, such as stem_pool and res1_pool, show broader responses that are consistent with coarse density patterns and local anatomical structure. In the deeper stages, the activations become smaller, sparser, and more selective. This progression is consistent with hierarchical feature learning in 3D CNNs, where the early layers capture general low-level structure and the later layers compress that information into more task-relevant representations. Even with only the benign example shown, the figure still makes the progression from broad spatial activation to more selective abstraction clear.

6. DISCUSSION

6.1. Task Interference was the Main Limitation

The clearest finding across the seven trials is that joint training on LUNA16 and NoduleMNIST3D consistently reduced performance on the main malignancy-classification task. Trial 4 showed the sharpest drop, with NoduleMNIST3D AUROC falling from 0.922 in Trial 3 to 0.830 after adding LUNA16 into the shared training pipeline. Trial 5 recovered only part of that loss, even after the LUNA16 preprocessing pipeline was brought closer to the MedMNIST normalization procedure. Performance only improved substantially once the model returned to single-task training in Trial 6, where AUROC rose to 0.915.

These results suggest that the main problem was not insufficient model size or weak optimization, but conflict between the objectives of the two tasks. LUNA16 nodule detection rewards the model for identifying a relatively coarse and visually obvious structure. NoduleMNIST3D malignancy classification is harder and depends on subtler cues such as margin irregularity, internal density variation, and fine-grained local structure. When both tasks were trained together, the shared backbone appears to have favoured features that were more useful for detection than for malignancy grading.

The two-phase strategy used in Trial 7 provided a more effective compromise. Pretraining on LUNA16 allowed the model to learn general 3D volumetric features, while fine-tuning on NoduleMNIST3D redirected the network toward the actual malignancy objective. This did not produce the highest AUROC in the study, but it improved the operating-point metrics, particularly F1, precision, specificity, and accuracy. That pattern suggests that LUNA16 can still be useful, but only when it is used for initialization rather than for simultaneous multi-task optimization.

6.2. Model Complexity Needed to Match the Task

The final results also showed that model capacity had to be matched to task difficulty. Deep3DCNN, especially with SE attention and two-phase training, was most appropriate for NoduleMNIST3D because malignancy classification depended on subtle, localized structural cues. In that setting, the stronger operating-point performance of Trial 7 suggests that the model benefited from richer feature extraction and channel reweighting, even though AUROC remained slightly below the Trial 6 peak.

By contrast, LUNA3DCNN reached AUROC 0.991 on LUNA16 without SE attention and without the larger capacity of Deep3DCNN. This is consistent with the nature of the detection task. The boundary between nodule and non-nodule patches in LUNA16 is much clearer than the boundary between benign and malignant nodules in NoduleMNIST3D. A smaller dedicated 3D model was therefore sufficient and, in practice, better suited to the problem.

A similar point applies to the chest X-ray task. CXRClassifier performed well as a separate 2D screening model because the data modality, feature scale, and clinical objective were different from those of the CT-based models. Keeping them separate avoided forcing one architecture to serve incompatible imaging domains.

6.3. Thresholds and Calibration were Informative Beyond AUROC

The threshold analysis provided useful information that AUROC alone could not capture. For Deep3DCNN, the selected threshold of 0.85 indicated that the best F1 region occurred at a relatively conservative operating point. This matched the confusion matrix, where false positives were reduced at the cost of some recall. For LUNA3DCNN, the chosen threshold of 0.55 sat near a naturally balanced region, which was reflected in the matched precision and recall of 0.934. For CXRClassifier, the calibrated threshold of 0.25 supported the intended screening behavior by maintaining recall at 0.997 while accepting lower specificity.

These differences reinforce the idea that the three tasks should not be treated as a single classification problem with one shared operating rule. Each model learned a different score distribution and required its own threshold depending on the clinical role of the task. In this study, AUROC remained useful as a threshold-independent summary of ranking quality, but the

threshold curves and confusion matrices were more informative when interpreting how each model would behave in practice.

6.4. Limitations

This study still has several limitations. First, the datasets are relatively small, especially for malignancy classification, which means that some trial-to-trial differences may be sensitive to split composition. Second, NoduleMNIST3D uses compact benchmark patches rather than full clinical CT volumes, so the model does not face the full localization burden of a real screening pipeline. Third, the chest X-ray dataset has a noticeable class-distribution shift between training and test splits, which likely affects calibration and specificity. Finally, the analysis in this work is based on retrospective benchmark evaluation rather than prospective clinical validation.

7. CONCLUSION

This work presented a seven-trial study of pulmonary nodule malignancy classification, nodule detection, and chest X-ray cancer screening using three dedicated deep learning models. The main conclusion is that task separation was more effective than shared multi-task training. When LUNA16 detection and NoduleMNIST3D malignancy classification were trained together, the shared backbone consistently underperformed on the main benchmark. Once the tasks were separated, performance improved substantially.

The final system consisted of Deep3DCNN for NoduleMNIST3D malignancy classification, LUNA3DCNN for LUNA16 nodule detection, and CXRClassifier for chest X-ray screening. Deep3DCNN achieved AUROC 0.906 with improved F1, precision, specificity, and accuracy at the tuned threshold. LUNA3DCNN achieved AUROC 0.991 with balanced precision and recall, showing that a compact dedicated detector was sufficient for the LUNA16 task. CXRClassifier achieved AUROC 0.889 and recall 0.997, which made it suitable as a high-sensitivity screening model despite its lower specificity.

Taken together, the results show that the strongest system was not the one with the most shared data or the most unified design. It was the one that respected the differences between tasks, imaging modalities, and decision requirements. LUNA16 pretraining remained useful when applied in a staged way, but not when combined with malignancy classification in a single optimization objective. More broadly, the study suggests that in medical imaging, good performance often comes from matching the model, data, and thresholding strategy to the exact clinical task rather than trying to force related problems into one shared framework.

Future work should examine higher-resolution CT patches, such as 48 × 48 × 48 or 64 × 64 × 64, to better preserve the subtle structural cues that separate benign from malignant nodules. It would also be useful to test more advanced transfer strategies, such as domain-adaptive or domain-adversarial methods, to determine whether some of the information in LUNA16 can be transferred without reintroducing the interference observed in the joint-training trials.

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